



Enzymolysis peptides from *Mauremys mutica* plastron improve the disorder of neurotransmitter system and facilitate sleep-promoting in the PCPA-induced insomnia mice

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ABSTRACT

Ethnopharmacology relevancy: For many centuries, *Mauremys mutica* is highly valued as a food homologous Chinese herbal medicine. It has been considered useful to sedate, nourish brain and promote sleep. However, the animal experimental evidence of its sleep-promoting activity is missing in literature.

Aim of the study: In this study, PCPA-induced insomnia model was used to explore the sleep-promoting mechanism of enzymolysis peptides from PMM, and its main composition and chemical structure were analyzed.

Materials and methods: Experiments were performed using PCPA-induced insomnia model, all animals were intraperitoneally injected with PCPA (350 mg/kg-d) for two days. The sleep-promoting effect evaluated using measuring content of 5-HT, GABA, DA, IL-1, BDNF and expression of 5-HT1A receptor and GABAA receptor $\alpha 1$ -subunit in mice brain. Primary structure of peptides was identified by HPLC-ESI-QqTOF-MS/MS.

Results: Compared with the model group, the content of 5-HT, GABA, IL-1, BDNF in mice brain of PMM peptide groups was increased to varying degrees, the content of DA was decreased, and the gene transcription and protein expression of 5-HT1A receptor and GABAA receptor $\alpha 1$ -subunit were almost all returned to normal levels. In addition, the primary structures of most abundant nine typical peptides in PMM peptides were identified.

Conclusions: The results showed that PMM peptides could improve the disorder of neurotransmitter system, restore compensatory over-expression 5-HT1A receptor and GABAA receptor $\alpha 1$ -subunit, and have a good sleep-promoting effect. The specific amino acid composition, sequence and glycosylation modification of PMM peptides may be the key reason for their activity, which lays a foundation for the subsequent development of sleep-promoting peptide products.

1. Introduction

Sleep is a physiological activity that plays an important role in human health. It helps energy storage and recuperation of the nervous system and can strengthen brain connections to help people learn and remember (Siegel, 2015). However, insomnia has increasingly become a common disease, with approximately 35% of people experiencing symptoms of insomnia throughout their lives (Mellinger et al., 1985). Lack of sleep can cause very serious health problems such as depression, heart disease, high blood pressure and obesity (Cunnington et al., 2013; Roth and Drake, 2004). BDZ are the most widely used sedative and hypnotic drugs and can induce sleep (Wang and Liu, 2016). Although

these drugs are beneficial, the side effects are very obvious, such as decreased cognitive function during the day, memory loss, and long-term use can lead to tolerance and addiction (Cuevas et al., 2003). For decades, sleep regulation has become a worldwide problem, hypnotic and sedative drugs with higher safety and efficacy are urgently needed. At the same time, traditional Chinese herbs with sedative and hypnotic effects are gradually accepted by the world for their safety and effectiveness (Gyllenhaal et al., 2000).

Due to the important role of bio-active peptides in promoting human health, the potential physiological functions of bio-active peptides extracted from proteins in traditional diet and medicinal materials have attracted extensive scientific interest and attention (Korhonen and Pihlanto, 2003; Mellander, 1951). At present, many reports have described

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Abbreviations

PMM	plastron of <i>Mauremys mutica</i>
PCPA	p-chlorophenylalanine
GABA	γ -aminobutyric acid
5-HT	5-hydroxytryptamine
DA	dopamine
BDNF	brain-derived neurotrophic factor
IL-1	interleukin-1
TFA	trifluoroacetic acid
qRT-PCR	Quantitative real-time polymerase chain reaction
GAPDH	glyceraldehyde-3-phosphate dehydrogenase
HPLC-ESI-QqTOF-MS/MS	high performance liquid chromatography electrospray ionization quadrupole time of flight mass spectrometry
BDZ	benzodiazepines
SDS-PAGE	sodium dodecyl sulfate-polyacrylamide gel electrophoresis
PDA	photo-diode array
PVDF	polyvinylidene fluoride
ECL	electro-chemiluminescence

the special biological activities of peptides, such as antibacterial, anti-tumor, antioxidant, immune enhancement, neuroregulation and neuroprotection (Miraliakbari and Shahidi, 2008; McLay et al., 2001; Wang et al., 2018). Among them, more and more attention has been paid to the effect of peptides on the higher functions of brain. Many polypeptides, especially those of aquatic animals, have been shown to play an important role in sleep, learning and memory. Marine collagen peptide has been proven to promote spatial memory and learning in mice by up regulating the expression of BDNF and PSD95 (Pei et al., 2010). The polypeptides obtained from the enzymatic hydrolysis of anchovies protein have been demonstrated to enhance memory by regulating acetylcholinesterase activity (Su et al., 2016). A peptide extracted from *Conus araneosus* also have been reported to have sleep-inducing effects in mice (Franklin and Rajesh, 2015). In addition, the absorption and utilization rate of peptides by human body is much faster than that of proteins and free amino acids (Hara et al., 1984). Enzymatic hydrolysis of proteins can rapidly produce a large number of bio-active peptide components, which are likely to become excellent health food or special medical food in the future.

Mauremys mutica, is a freshwater turtle, distributed in Vietnam and south-central China (Cheng et al., 2010). It is highly valued as a kind of medicine or food homologous Chinese herbal and has been developed for centuries as functional foods. Both Guang Dong Sheng Zhi (Yi Yao Zhi) and Pharmacopoeia of the People's Republic of China regard turtle plastron as a kind of precious medicinal material. *Compendium of Materia Medica* records that plastron of *Mauremys mutica* (PMM) have the functions of detoxification, anti-inflammation, sedation and brain benefiting. Studies have shown that some turtle extracts have good antioxidant, antihypertensive and antimicrobial effects (Liu et al., 2012; Chiu et al., 2006). In recent years, people pay more and more attention to the research of *Mauremys mutica*. The antioxidant and immune system enhancement effects of *Mauremys mutica* protein and peptide extracts in mice have been confirmed (Yin et al., 2016; Lv et al., 2019). However, we are still not sure whether the PMM peptides have a certain effect on promoting sleep and protecting the nervous system. Oxidative stress can lead to cell and tissue damage, and antioxidant can protect neurotransmitters and sleep-related cytokines in the brain from damage. In addition, there are many endogenous peptides that promote sleep in human brain (McLay et al., 2001). We have reason to assume that the PMM peptides have some physiological functions of promoting sleep. The purpose of this experiment is to explore the sleep promoting effect of

PMM polypeptides by establishing the PCPA-induced insomnia model in mice. We detected the expression of sleep-related cytokines, neurotransmitters and their receptors in brain tissues to prove the regulatory effect of PMM peptides in the brain, and further revealed its molecular mechanism of promoting sleep.

High performance liquid chromatography electrospray quadrupole time of flight mass spectrometry (HPLC-ESI-QqTOF-MS/MS) is an effective tool for identification and analysis of bioactive substances, especially polypeptide sequences (Cui et al., 2000; Medzihradsky et al., 2001). According to previous literature, 48 peptides with antitumor activity in corn were identified by HPLC-ESI-MS/MS system (Wang et al., 2016), and the primary structure of antimicrobial peptides in *Phyllomedusa hypochondrialis* skin secretion were analyzed by RP-HPLC and MS/MS (Huang et al., 2017). In this study, the primary structure of PMM peptides was identified by HPLC-ESI-QqTOF-MS/MS. The differences of peptide component concentration and amino acid sequence in structure identification are helpful to reveal the relationship between peptide structure and physiological function. Based on the experimental results, it is helpful to find the key components of active peptide for sedation and hypnosis, reveal the physiological mechanism of active peptide, and lay the experimental foundation for the development of PMM high-value products.

2. Materials and methods

2.1. Materials and reagents

The PMM peptides are derived from plastron of *Mauremys mutica* donated by Dongguan Shengui Biotechnology Co., Ltd. (Guangdong, China) in September 2018. The *Mauremys mutica* were authenticated by Yang-yan Jiang, vice-chairman of the Tortoises and Turtles Association of Dongguan City. The voucher specimens of *Mauremys mutica* (No.180920) have been deposited in College of Food Science and Technology, Huazhong Agricultural University. PCPA was purchased from Shanghai Yien Chemical Technology Co., Ltd (Shanghai, China). HPLC-grade acetonitrile was bought from Fisher Chemical (Fair Lawn, NJ, USA), and HPLC-grade water was prepared using a Milli-Q system (Millipore Ibérica, Madrid, Spain). Trifluoroacetic acid (TFA) of HPLC grade was purchased from Sigma Chemical Co., Ltd. (Saint Louis, MO, USA). Assay kits for the determination of 5-HT, DA, GABA, BDNF, IL-1 were bought from Nanjing Jiancheng Biotechnology Co., Ltd. (Nanjing, China). RIPA Lysis Buffer, BCA protein quantitative detection kit and SDS-PAGE gel preparation kit were purchased from Servicebio Technology Co., Ltd (Wuhan, China). All other chemicals and reagents were of analytical purity.

2.2. Preparation and main compositions of PMM peptides

The *Mauremys mutica* was killed in hot water, and its plastron was collected and cleaned. The extraction of PMM protein and the further preparation of polypeptide refer to the previous research in our laboratory (Lv et al., 2019; Yin et al., 2016). The plastron was boiled twice in distilled water of eight times the mass for 3 h each time. The liquid is then evaporated and concentrated at 55 °C until the condensate is no longer flowing out. After the liposoluble components were removed by petroleum ether extraction, the concentrate was put into the vacuum freeze-drying equipment (China Beijing Songyuan Huaxing Technology Development Co., Ltd.) for 48 h, and the total extract was obtained by crushing.

The total extracts were weighed 5 g in a 250 mL conical flask and added pH 7.5 NaOH solution at a solid-liquid ratio of 1:25 (g/mL). And 1.2% alkaline protease (60U/mg) was used to hydrolyze protein extract. The reaction system was heated in water bath at pH 7.5 and 55 °C for 6 h, and then boiling water bath for 10 min immediately to inactivate the enzyme. After centrifugation at 4 °C, 10,000 r/min for 10 min, the supernatant was taken to obtain the enzymatic hydrolysate, then

transferred to an ultrafiltration system ($M_w < 5$ kDa) (Millipore Co., Bedford, MA, USA). PMM peptides were finally prepared by rotary evaporation concentration and freeze-drying.

The method recommended by the standardization management committee of the People's Republic of China (GB/T 22,492-2008, GB 5009.6-2016, GB 28050-2011, GB 5009.3-2016) was used to determine the content of nutrients in freeze-dried powder. The results show that the freeze-dried powder mainly contains polypeptides (72.16%), and the determination results of various nutrients are shown in Table S1.

2.3. HPLC analysis

PMM peptides were purified by adding ethanol, TFA and sodium chloride, then filtered, and prepared into 10 mg/mL peptide solution. Shimadzu LC-20A HPLC-PDA detection system (Shimadzu Co., Ltd, Kyoto, Japan) was used to HPLC analysis. The separations of the peptide solution were carried out on Inter Sustain AQ-C18 column (250 mm $\times$ 4.6 mm, 5 μ m) at a flow rate of 0.8 mL/min. The temperature of the column was maintained at 30 °C. The mobile phases were consisted of ultrapure water (with 0.1% TFA, A) and acetonitrile (with 0.1% TFA, B). A linear gradient program was optimized as follows: 0–9 min, 100 to 91% A; 9–12 min, 91 to 87% A; 12–20 min, 87 to 74% A; 20–60 min, 74 to 30% A. After each run, an equilibration time of 20 min was set. The injection volume was 8 μ L and the detection wavelength was 215 nm.

2.4. HPLC-ESI-QqTOF-MS/MS analysis

PMM peptides were analyzed using HPLC-ESI-QqTOF-MS/MS system (AB SCIEX Co., Ltd, Shanghai, China) and the column and gradient elution system was consistent with those used in HPLC-PAD analysis. Mass spectrometry conditions were as follows: ESI + ion source; nebulizer, 40.00 psi; capillary voltage, 3500 V; dry gas flow rate, 10.0 L/min; drying gas temperature, 325 °C; scan spectra from m/z 100 to 1500.

The MS/MS spectra were analyzed by using the software PeakView 2.2. The cleavage pattern of peptide chain was shown in Fig. S1. The b series ions and y series ions formed by peptide bond cleavage were the most common ions in mass spectrometry and had relatively high signal intensity. Therefore, the primary structure identification of the peptide chain was mainly based on y and b fragment ions.

2.5. Animal administration

Male Kun-Ming strain mice (male, SPF, 20 ± 2 g, 8 weeks old) were purchased from Huazhong Agricultural University (Certificate No. SCXK (Hubei) 2015-0019, Wuhan, China) and used in this study. Under a normal light/dark (12 h/12 h) cycle, all mice were housed at a constant temperature of 25 ± 1 °C with a relative humidity of $60 \pm 10\%$. During the study period, animals were given free access to standard laboratory granular diets and distilled water. In addition, all studies were conducted in strict accordance with the standards for laboratory animals established by the People's Republic of China (GB14925-2001) and the Declaration of Helsinki. The animal protocol was approved by the Scientific Ethics Committee of Center for Laboratory Animals in Huazhong Agricultural University (Permission No. HZAUMO-2019-017).

2.6. Suprathreshold dose of pentobarbital sodium sleep experiment

In the experiment, mice were randomly divided into blank control group, low-dose PMM peptide group and high-dose PMM peptide group, 10 mice in each group. The control group was given 0.1 mL/(10 g d) normal saline, low-dose PMM peptide group and high-dose PMM peptide group were given 400 mg/(kg $\cdot$ d) and 800 mg/(kg $\cdot$ d) peptide solution for twelve days. On the 12th day, each group was given intraperitoneally with suprathreshold dose of pentobarbital sodium (45 mg/kg, 0.1 mL/10 g). Then the number of mice with the disappearance of righting reflex after a period of time was recorded, as well as the sleep

number, sleep time and awakening time (Li et al., 2018).

2.7. Animal model of PCPA-induced insomnia in mice

The mice were randomly divided into 5 groups, 10 mice in each group, including blank control group, model group, diazepam group, low-dose PMM peptide group, high-dose PMM peptide group. After four days of adaptive feeding, the control group was given 0.1 mL/(10 g d) normal saline by gavage, and the other groups were given 350 mg/(kg $\cdot$ d) PCPA by intraperitoneal injection for two days. From the 4th day, the model group was given 0.1 mL/(10 g d) saline, diazepam group was given 1.2 mg/(kg $\cdot$ d) diazepam solution, low-dose PMM peptide group and high-dose PMM peptide group were given 0.5 g/(kg $\cdot$ d) and 1 g/(kg $\cdot$ d) peptide solution, respectively, for 8 days, to observe the recovery of mice in each group. On the 14th day, 2 h after the administration, the mice in each group were intraperitoneally injected with pentobarbital sodium 45 mg/(kg $\cdot$ d), and the number of sleeping mice, sleep latency and sleep time were recorded to evaluate the modeling.

2.8. Determination of organ index and neurotransmitter, IL-1, BDNF in brain

After administration on the 13th day, the mice were deprived of food but not water for 12 h. Then, the mice were killed an hour after the last administration on the 14th day. After the blood was taken from the orbital area of the mice, the mice were killed, and the brain, liver and kidney tissues were rapidly and carefully separated on the ice plate, washed with normal saline, and weighed one by one quickly.

Organ index was the percentage of the ratio of organ mass to animal mass. Its index could reflect the health status of animal organs. The calculation formula was as follows:

Brain tissue was added 0.9% normal saline to make 10% homogenate, centrifuged in a 4 °C frozen centrifuge for 15 min at 4500 r/min. Then the upper layer was extracted overnight and stored in a clean centrifuge tube for record numbering and freezing. The contents of 5-hydroxytryptamine (5-HT), dopamine (DA), γ -aminobutyric acid (GABA), interleukin-1 (IL-1) and brain-derived neurotrophic factor (BDNF) were determined by ELISA kit.

2.9. qRT-PCR analysis

The brain tissue of each group of mice was crushed with Trizol reagent, and then centrifuged at 4 °C and 12,000 r/min for 10 min to obtain the supernatant. Total RNA was extracted by adding chloroform and isopropanol in order, and the concentration and purity of RNA were determined by Nanodrop 2000 (Thermo Fisher Scientific Co., Ltd, Shanghai, China). Total RNA was added to a PCR tube, and then reverse transcribed into cDNA using a reverse transcription kit. All operations were performed on ice. The reaction system was configured for amplification. PCR amplification reaction system: 2 $\times$ qPCR Mix, 12.5 μ L; 7.5 M gene primer, 2.0 μ L; reverse transcription product, 2.5 μ L; ddH₂O, 8.0 μ L. The target gene and the internal reference gene were amplified simultaneously under the same reaction conditions. Reaction conditions: Predenaturation, 95 °C, 10 min; 40 cycles (95 °C, 15 s; 60 °C, 60 s). After 40 cycles, the dissolution curve was analyzed and qRT-PCR was performed with StepOnePlus RT-PCR system (Thermo Fisher Scientific Co., Ltd, Shanghai, China). The 5-HT_{1A} receptor and GABA α 1-subunit mRNA were calculated using β -actin and GAPDH as internal parameters, and the relative expression of the product was calculated using $2^{-\Delta\Delta Ct}$. The internal control gene and target gene primer sequences are shown in Table S2.

2.10. Western blot analysis

Protein extraction: total protein extraction kit was used to extract the total protein of brain tissue. The brain tissue of mice was mixed with

precooled RIPA buffer, homogenized by ultrasonic homogenizer, and then put into shaker and incubated in ice water bath for 30 min. After incubation, it was centrifuged at 12,000 r/min and 4 °C for 10 min to obtain the supernatant which is the total protein sample of tissue. Then the protein samples were quantitatively analyzed by BCA kit, and the protein solution was added into 5 × protein sample loading buffer in the proportion of 4:1, and denatured in boiling water bath for 15 min.

SDS-PAGE: the SDS-PAGE gel was prepared with a 10% separation gel and a 5% concentrated gel. Put the gel maker into the electrophoresis tank with enough electrophoresis solution, and add the sample into the sample hole for electrophoresis. The voltage of concentrated gel is 75 V, and that of separation gel is 120 V. When bromophenol blue just ran out, the electrophoresis was stopped and the membrane was transferred.

Western blotting: PVDF membrane immersed in anhydrous methanol was transferred to buffer in ice bath. After transfer, PVDF membrane was incubated in TBST blocking buffer containing 5% skimmed milk powder for 1 h. The main antibody of the protein was diluted with TBST containing 5% skimmed milk powder, incubated overnight at 4 °C, and then added with appropriate dilutions of primary antibody and secondary antibody to make it react with the target completely. After the PVDF film was dried, the ECL chromogenic agent was soaked for 5 min for western blotting. Imaging, fixing and film processing are all carried out in a dark room. AlphaEaseFC software processing system is used to analyze the optical density value of the target band, and beta-actin is used to represent the gray ratio of the target band.

2.11. Statistical analysis

The results were expressed in the form of mean ± standard and evaluated by one-way ANOVA of IBM SPSS Statistics 20 for windows. Duncan's multiple comparisons were used to analyze the differences between groups, and the statistically significant level was $P < 0.05$.

3. Results and discussion

3.1. Effect of PMM peptides on sleep, body weight and organ index of mice in pentobarbital sodium hypnosis experiment

The method of intraperitoneal injection of pentobarbital sodium is used to make mice sleep. The selection of subthreshold dose can ensure that almost all mice fall asleep, so as to more intuitively measure their sleep time and other changes, and better study the sleep promoting effect (Chu et al., 2007; Ma et al., 2009). As we can see in Table 1, all mice fell asleep after hypnosis with pentobarbital sodium. Compared to the control group, PMM peptide groups both produced a significant decrease in sleep latency and a significant increase in sleep time. These results indicated that PMM peptide has a good sleep promoting effect. The body weight change and organ index of mice in each group were shown in Fig. 1 and Table S3. PMM peptide groups had no significant difference in weight and organ index compared to the control group, indicating that PMM peptides had little effect on the health of normal mice after 12 days of administration.

Table 1

The effect of PMM peptide on sleep latency and sleep time of normal mice ($\bar{x} \pm S$) ($n = 10$). Different letters marked above the bar are significantly different by ANOVA multiple test ($P < 0.05$).

Group	Sleep rate of mice	Sleep latency (min)	Sleep time (min)
Control	100%	6.805 ± 3.132^a	39.18 ± 7.530^c
Low-dose PMM peptide	100%	5.153 ± 0.353^b	69.30 ± 8.860^b
High-dose PMM peptide	100%	4.363 ± 0.503^c	89.86 ± 8.631^a

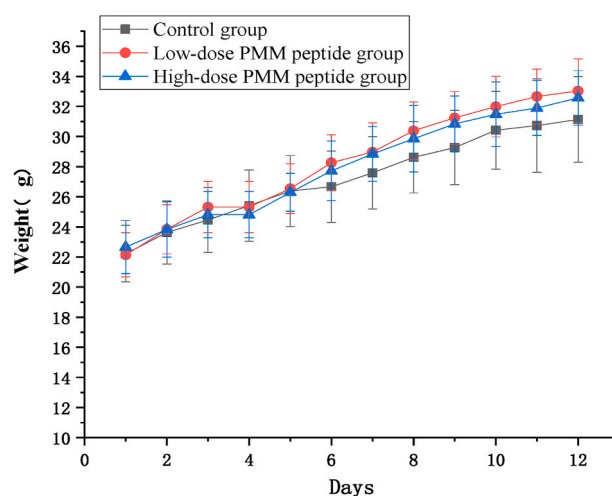


Fig. 1. The effects of PMM peptides on weight changes in hypnotic mice ($\bar{x} \pm S$) ($n = 10$). There was no statistical difference between the other groups and the model group, so there was no annotation.

3.2. Effect of PMM peptides on sleep, body weight and organ index of mice in PCPA model

The PCPA model with the characteristics of high success rate and typical pathological changes is a more recognized insomnia model in domestic and international. PCPA can significantly reduce the 5-HT content in the brain by inhibiting tryptophan hydroxylation and blocking the synthesis of 5-HT, thus causing anxiety, insomnia and other symptoms (Borbély et al., 1981; Jouvet, 1999). In this experiment, after two days of intraperitoneal PCPA injection, mice lost their circadian rhythm and became irritable. The sleep of all mice was shown in Table 2. Compared with the control group, the sleep time of the mice in the model group decreased significantly, and the sleep latency became significantly longer, indicating that modeling was successful. Compared with the model group, the PMM group all showed a significant larger increase in sleep time, and the sleep latency was significantly reduced. The sleep quality of the mice in the low-dose PMM peptide group was similar to that in the blank group, and the sleep quality of the mice in the high-dose PMM peptide group was better than that in the low-dose group. The results showed that PMM peptides had a good effect on the recovery of insomnia in mice induced by PCPA, and high dose of PMM peptides had a hypnotic effect nearly comparable to that of diazepam.

The body weight change and organ index of mice in each group were shown in Fig. 2 and Table S4. There was no significant difference in liver index and kidney index among all groups. However, compared to the control group, the weight of other groups decreased significantly after establishing insomnia model for two consecutive days from the 5th day. On the 7th day, the mice were intragastrical administered, and the body weight of mice in the diazepam group and the PMM peptide groups began to approach the control group. And on the 12th day, there was a

Table 2

The effect of PMM peptide on sleep latency and sleep time of PCPA-induced insomnia mice ($\bar{x} \pm S$) ($n = 10$). Different letters marked above the bar are significantly different by ANOVA multiple test ($P < 0.05$).

Group	Sleep rate of mice	Sleep latency (min)	Sleep time (min)
Control	100%	5.259 ± 0.695^b	35.43 ± 6.921^c
Model	80%	8.740 ± 0.979^a	20.57 ± 4.906^d
Diazepam	100%	3.976 ± 0.589^c	70.49 ± 12.88^a
Low-dose PMM peptide	100%	5.027 ± 0.391^b	38.52 ± 7.683^c
High-dose PMM peptide	90%	4.596 ± 0.767^{bc}	55.40 ± 8.104^b

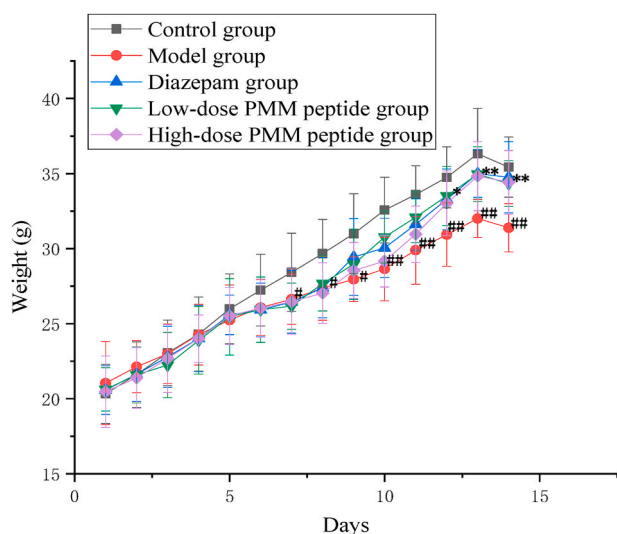


Fig. 2. The effects of PMM peptides on weight changes in PCPA-induced insomnia mice ($\bar{x} \pm S$) ($n = 10$). In the comparison between model group and control group, # means $P < 0.05$, and ## means $P < 0.01$. And in the comparison between the model group and each drug administration group (diazepam group and PMM peptide groups), * indicates $P < 0.05$, and ** indicates $P < 0.01$. There was no statistical difference between the other groups and the model group, so there was no annotation.

significant difference between the model group and administration group. On the last day of the experiment, the body weight of the diazepam group and the PMM peptide groups had no significant difference from the control group, indicating that both diazepam and PMM peptides had a good effect on recovering the weight loss caused by insomnia in mice.

3.3. Effect of PMM peptides on 5-HT, DA, GABA, IL-1, BDNF levels in brain tissues

Sleep/wake is a complex physiological process that is regulated by the activity of multiple parts of the brain, among which the neurotransmitter system plays a very important role (Nishino, 2013; Yan et al., 2015). The most closely related neurotransmitters to sleep are: 5-HT, DA, GABA, which should be detected in all mice in this experiment. GABA and 5-HT are two of the most important sleep promoting neurotransmitters in the brain, which can reduce the activity of neurons and regulate the function of nerve cells. However, DA usually conducts excitement and nerve impulses, which is related to wakefulness (Li and Jasanoff, 2020). As shown in Fig. 3, compared to the control group, the content of 5-HT and GABA in the model group mice decreased significantly, and the content of DA increased significantly, indicating that after PCPA induction, not only the synthesis of 5-HT was specifically blocked, but also the disorder of neurotransmitter system was caused. Compared to the model group, low- and high-dose PMM peptide group produced a significant increase in 5-HT and GABA content, and the high-dose group was optimal, nearly comparable to the diazepam effect. Compared with the model group, the content of DA in low-dose PMM peptide group decreased significantly, but the high-dose group was not very obvious. The results show that PMM peptides has a certain recovery effect on neurotransmitter system, and its sleep promoting effect may be mainly achieved by regulating the contents of 5-HT and GABA neurotransmitters.

IL-1 is a kind of cytokine with immunoregulatory effect, which plays an important role in host defense, and is also the earliest discovered cytokine with sleep-inducing effect (Imeri and Opp, 2009; Krueger et al., 2011). Previous studies have shown that IL-1 injection can increase non-rapid eye movement sleep (Manfridi et al., 2003). As shown in Fig. 3D, there was no significant difference in IL-1 content in all groups except high-dose PMM peptide group. Compared to other groups, the

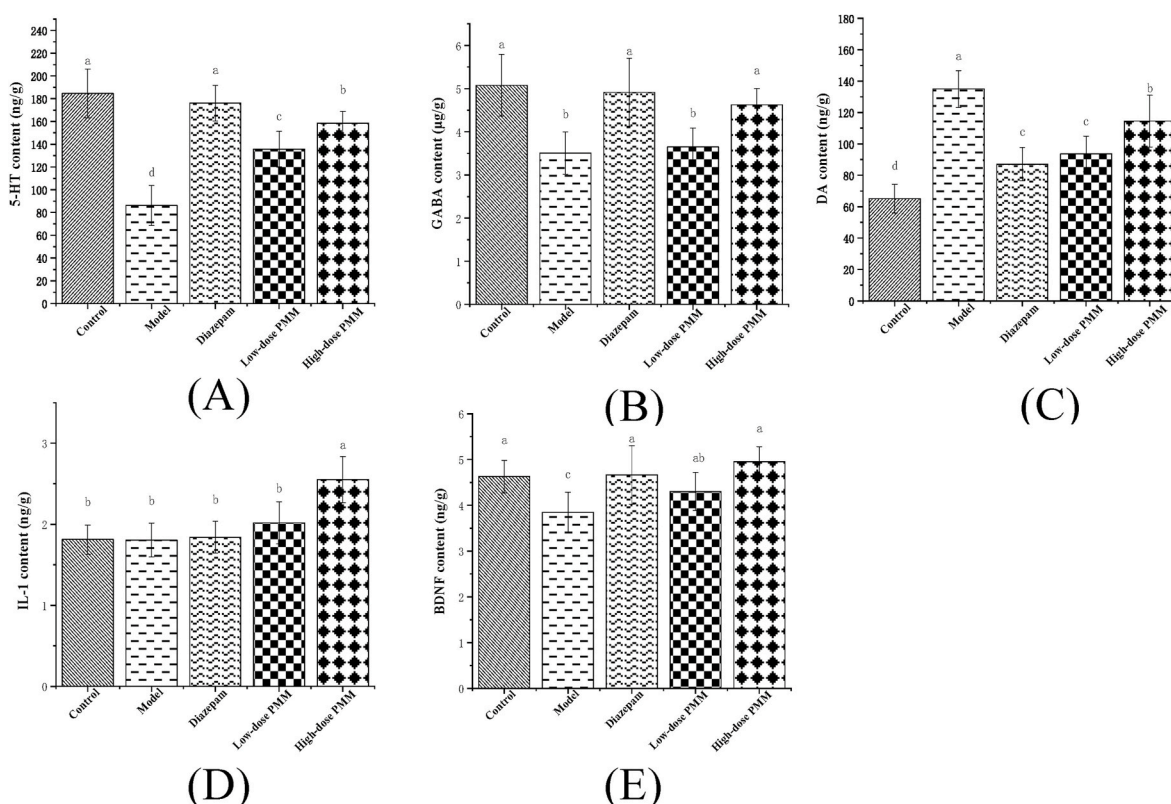


Fig. 3. The effects of PMM peptides on 5-HT (A), GABA (B), DA (C), IL-1 (D) and BDNF (E) levels in brain of PCPA-induced insomnia mice. Values expressed as the $\bar{x} \pm S$ ($n = 10$). Different letters marked above the bar are significantly different by ANOVA multiple test ($P < 0.05$).

high-dose PMM peptide group produced a significant increase, indicating that PMM peptide may cause the rise of IL-1 and regulate sleep. It has been shown that both protein and peptide extract of PMM can enhance the immune function of mice to a certain extent (Lv et al., 2019), and immune activation can participate in sleep-wake behavior regulation network through cytokines (Krueger et al., 2011), which may be one of the mechanisms of PMM to promote sleep. BDNF contributes well to neurodevelopment, synaptic formation, neuroplasticity, and sleep regulation (Pillai and Buckley, 2012; Nieto et al., 2013). In addition, it can maintain and promote the growth and regeneration of 5-HT, DA and GABAergic neurons, and help to protect the brain neurotransmitter system. The BDNF content in each group of mice was shown in Fig. 3E. Compared to the control group, model group mice produced a significant decrease, indicating that BDNF may be degraded during insomnia and lead to more serious consequences such as cognitive disorders. PMM peptide groups all produced a significant increase in BDNF compared to the model group, and the high-dose of PMM peptides restored the BDNF content in mice brain to be consistent with the control group and diazepam group. Taken together, the results show that PMM peptides can play a good role in promoting sleep by increasing the content of IL-1 and BDNF cytokines, and promote the protection and recuperation of nervous system damage in insomnia.

3.4. Effects of PMM peptides on mRNA and protein levels

The 5-HT_{1A} receptor and GABA_A receptor α 1-subunit are essential for regulating the neurotransmitter system in the brain. Their deletion or over-expression may lead to mental disorders, such as depression, anxiety, insomnia or absence epilepsy (Wang et al., 2019; Richardson-Jones et al., 2010; Seo and Leitch, 2015; Kyungae et al., 2018). After induced insomnia by PCPA, both 5-HT and GABA levels were significantly reduced in mice, and as shown in Fig. 4, their corresponding receptors showed compensatory over-expression ($P < 0.05$), which was also consistent with previous studies when neurotransmitters were scarce (Feng, 2014; Gutknecht et al., 2012; Kim et al., 2014). Over-expression of GABA_A receptor α 1-subunit may cause anxiety (Salari et al., 2015), while over-expression of 5-HT_{1A} receptor and reduction of 5-HT may be one of the possible causes of depression (Wang et al., 2019; Richardson-Jones et al., 2010). This suggests that after a period of insomnia, it

may affect the expression of receptor and lead to more serious mental illness. The change of GABA_A receptor α 1-subunit mRNA expression in each group is shown in Fig. 4A. Except for the model group, there is no significant difference between each group and the blank group, indicating that PMM peptides may regulate the over-expression of this receptor, and the abnormality of this receptor may be easy to recover. The change of 5-HT_{1A} receptor mRNA expression in each group is shown in Fig. 4B. Compared with the model group, PMM peptide groups both have a significant decrease ($P < 0.05$), while high-dose PMM peptide group has no significant difference from the blank group. However, diazepam group failed to improve the abnormal expression of 5-HT_{1A} receptor mRNA. Then we further measured the protein expression of 5-HT_{1A} receptor with western blotting in Fig. 4C, and the model group still produced a significant increase compared with the blank group. There was no significant difference between the low-dose PMM peptide group and diazepam group and the model group, while the high-dose PMM peptide group decreased the expression of 5-HT_{1A} receptor protein to the same level as the blank group ($P > 0.05$). The results showed that PMM peptides may reduce the more serious mental diseases caused by long-term insomnia by improving the over-expression of 5-HT_{1A} receptor and GABA_A α 1-subunit receptor.

3.5. Primary structure identification of PMM peptides

Biological activities of protein hydrolysates are related to amino acid composition, sequence, size and configuration of peptides (Matsui et al., 2002). Studies have shown that the molecular weight of polypeptides is closely related to their biological activities, and the low molecular weight polypeptides composed of 2–9 amino acids often have a good antioxidant effect (Lin et al., 2011). Therefore, the scanning range of mass spectrometry was set at m/z 50–1500, and other mass spectroscopic conditions have been proved to be useful for the analysis of PMM peptides in our previous studies (Lv et al., 2019). There are five peaks in the total ion chromatograms shown in Fig. 5A, among which the fourth peak with retention time of 3.243 has the largest component content. ESI-MS and MS/MS analysis were carried out by using PeakView 2.2 analysis software. The structure of PMM polypeptide was determined and 9 peptide segments were identified. The specific peptide information is shown in Table 3.

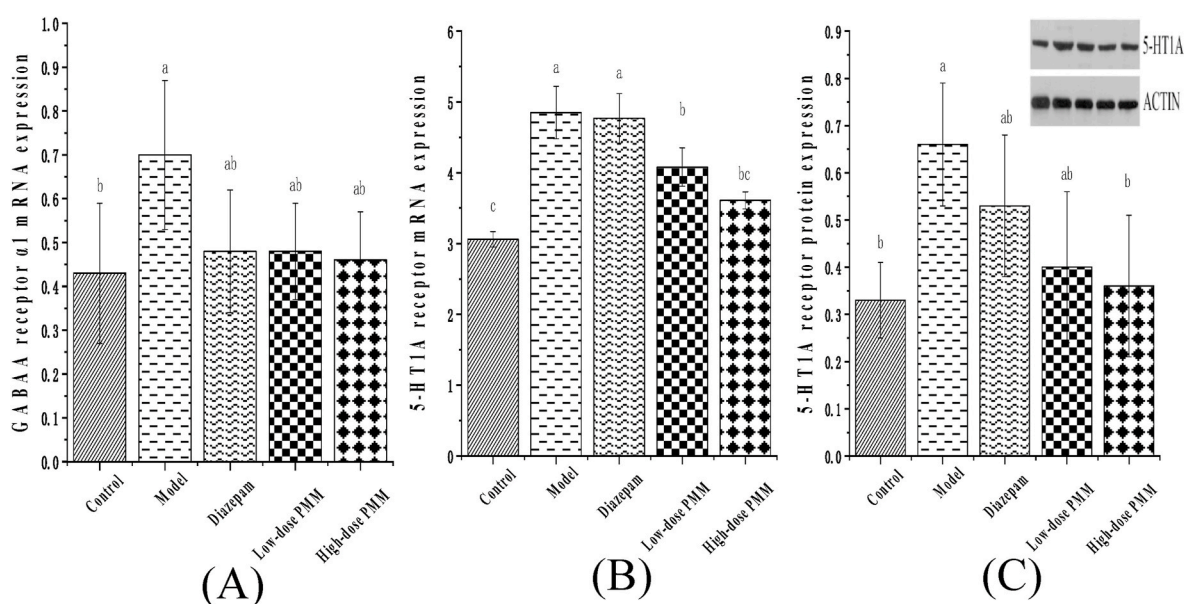


Fig. 4. The effects of PMM peptides on mRNA and protein expression levels. (A) Comparison of GABA_A receptor α 1-subunit mRNA expression in each group. (B) Comparison of 5-HT_{1A} receptor mRNA expression in each group. (C) Comparison of 5-HT_{1A} receptor protein expression in each group. Different letters marked above the bar are significantly different by ANOVA multiple test ($P < 0.05$).

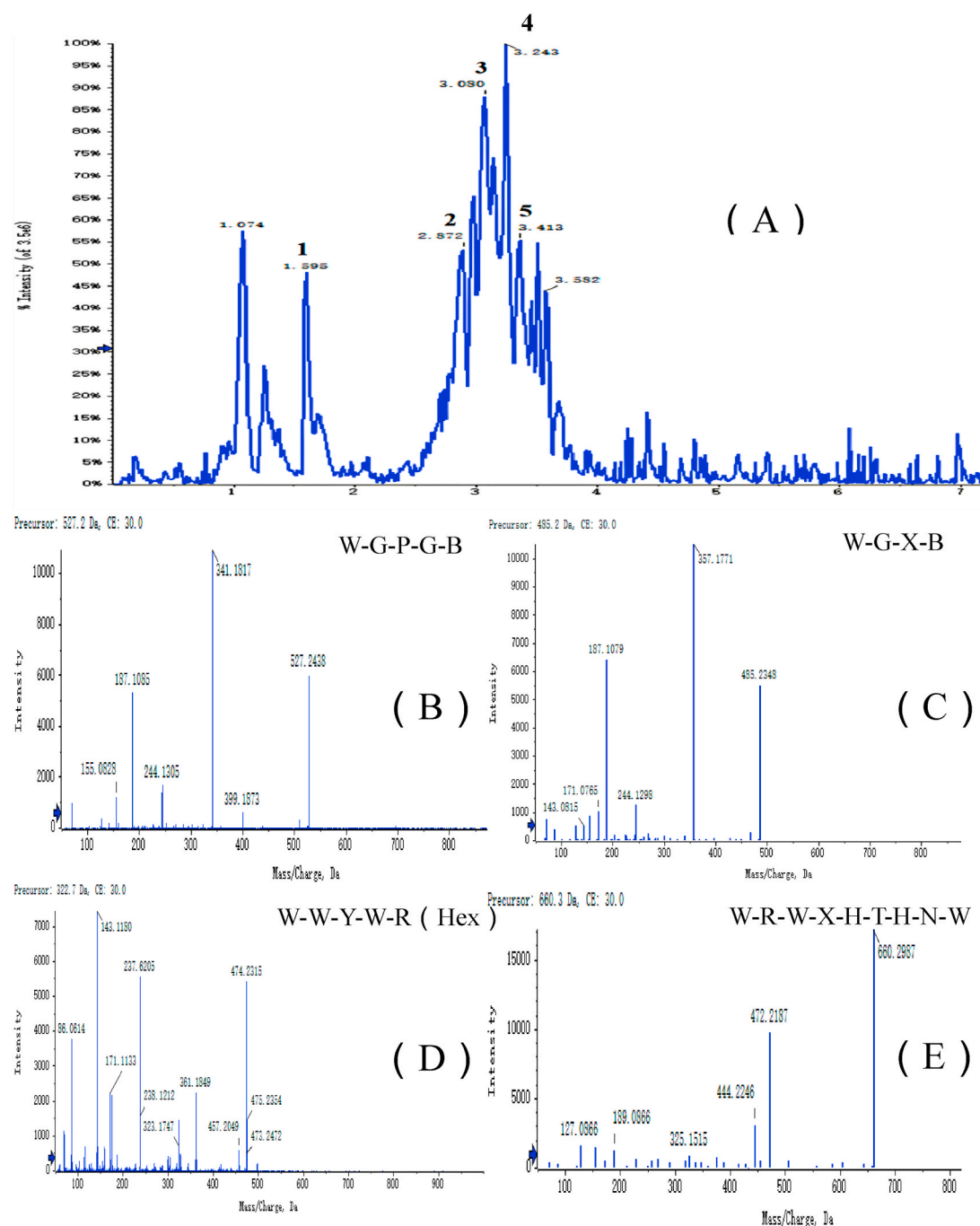


Fig. 5. (A) Total ion chromatograms of PMM peptides. (B, C) Two typical fragments of peak 2 in HPLC-ESI-QqTOF-MS/MS identification of PMM peptides, the structure of which are W-G-P-G-B and W-G-X-B. (D, E) HPLC-ESI-QqTOF-MS/MS of PMM peptides in peak 3 and peak 4, the structure of which are W-W-Y-W-R (Hex) and W-R-W-X-H-T-H-N-W. Hex = Hexose glycosylation modification.

Peak 2 consists of three main components (m/z 485.2348, 527.2438 and 415.2403), among which the MS2 spectrum of one component is used to illustrate the structural identification process. For example, the fragment of m/z 527.2438, which was the longest chain among them. As is shown in Fig. 5B, the m/z 527.2438 and 509.2336 were the quasi-molecular ion peaks $[M+H]^+$ and $[M-OH+H]^+$. After calculation, m/z 510.2336 - m/z 399.1873 = 111.0463, m/z 399.1873 (b4) - m/z 341.1817 (b3) - 1.0073 (H) = 56.9983, $b3-b2$ = 97.0512, $b2-b1$ = 57.0220 and $b1$ = 187.1085. The relative molecular weight shown in the above calculation results corresponds to five kinds of amino acids: Pyr (b), Gly (g), pro (P), Gly (g) and Trp (W). Therefore, we can deduce that the amino acid sequence from N-terminal to C-terminal is W-G-P-G-B,

and the primary structure of the peptide of this major component was identified.

According to the identification results in Table 3, we found that many amino acids of PMM peptides have been proved to have hypnotic and sedative effects, such as Ala (a), Arg (R), Asn (N), Gln (Q), Gly (G), Pro (P), Trp (W), Hyp (X), etc (Furuse, and Mitsuhiro, 2015). Peptides have higher absorption efficiency than free amino acids, which may play a better effect. In addition, there is evidence that the antioxidant amino acids Pro (P), Ala (A), His (H), etc., as effective proton/hydrogen donors, play an important role in the antioxidant activity of peptides (Wong et al., 2020). The N-terminal has hydrophobic amino acids which may also enhance the antioxidant function of peptides (Nwachukwu and

Table 3

The identification results of PMM peptides were analyzed by HPLC-ESI-QqTOF-MS/MS.

peak	t _R (min)	theoretical mass (m/z)	fragment ions(m/z)	N→C
1	1.591	232.1407	215.1135/158.0927	R-G
	1.599	329.1916	312.1667/175.1187	J-H
	1.602	542.2726	524.2479/414.2052/ 301.1514/244.1299/ 116.0705	N-Q-G-X-B
2	2.865	485.2348	468.2713/357.1771/ 244.1298/187.1079	W-G-X-B
	2.872	527.2438	510.2336/399.1873/ 341.1817/244.1305/ 187.1085	W-G-P-G-B
	2.893	415.2403	398.2284/301.1876/ 244.1658/130.0863	K-N-G-P
3	3.033	1372.6586	458.2236/440.2154/ 244.1296/187.1082/ 155.0820/127.0807	W-W-Y-W-R- (Hex)
4	3.243	1338.6778	879.4575/778.4096/ 669.3389/660.2986/ 642.3077/566.8095/ 529.2630/511.2505/ 344.2020/187.2247	W-R-W-X-H- T-H-N-W
5	3.415	1118.6152	559.3076/541.2988/ 454.2669/341.1836/ 284.1618/188.1040/ 171.0770/86.0973	W-N-A-R-W- P-G-J

Aluko, 2019). There are many antioxidative amino acids in the peptide of peak 2–5, and there are hydrophobic amino acids Trp (W) in the N-terminal of many peptide segments, which may be related to the protection of neurotransmitters and cytokines in the brain by the peptide through its antioxidant function. The repeated amino acid sequence and hexosylation modification in the peak 3 peptide have been proved to be related to the high biological value of the peptide (Siow and Gan, 2013; Arihara et al., 2017). In general, based on the relationship between the structure and activity of the identified PMM peptides, it is reasonable that the identified peptides may be able to exert a good sleep-promoting effect.

4. Conclusions

In conclusion, this study successfully verified the sleep-promoting effect of PMM peptides through animal experiments *in vivo*, and identified the primary structure of the main peptide components. In the experiment, high-dose PMM peptides showed the best effect in promoting sleep, improving the disorder of neurotransmitter system, increasing the content of sleep promoting factors and normalizing the expression of 5-HT_{1A} receptor and GABA_A receptor α 1-subunit. Diazepam, a positive drug, was superior to PMM peptides in increasing the contents of 5-HT and GABA, but failed to regulate the expression of neurotransmitter receptor. This also shows that PMM peptides not only improve sleep, but also may prevent insomnia from causing more serious mental illness. In the final structure identification process, we found that many amino acids of PMM peptides have sedative and anti-oxidation effects, and the repeated amino acid structure and hexosylation modification of individual peptide segments provide evidence for the high activity value of the peptides. This study provides a basis for the high-value utilization of *Mauremys mutica*, established a good experimental basis for exploring functional food and medical food of sleep-promoting function of polypeptides.

Author contributions

Yan-bo Lv: Conceptualization, Methodology, Investigation, Software, Writing – original draft. Qing Zhou: Methodology, Resources, Project administration, Funding acquisition. Jia-xing Yan: Methodology,

Resources. Lin-song Luo: Methodology, Resources. Jiu-liang Zhang: Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2021.114047>.

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